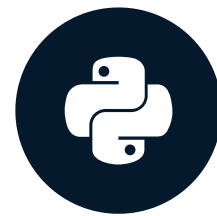


# Utilizar pandas con Seaborn

INTRODUCCIÓN A LA VISUALIZACIÓN DE DATOS CON SEABORN



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# ¿Qué son los pandas?

- Biblioteca Python para análisis de datos
- Lee fácilmente conjuntos de datos de archivos csv, txt y otros tipos de archivos.
- Los conjuntos de datos adoptan la forma de objetos `DataFrame`

# Trabajar con DataFrames

```
import pandas as pd
df = pd.read_csv("masculinity.csv")
df.head()
```

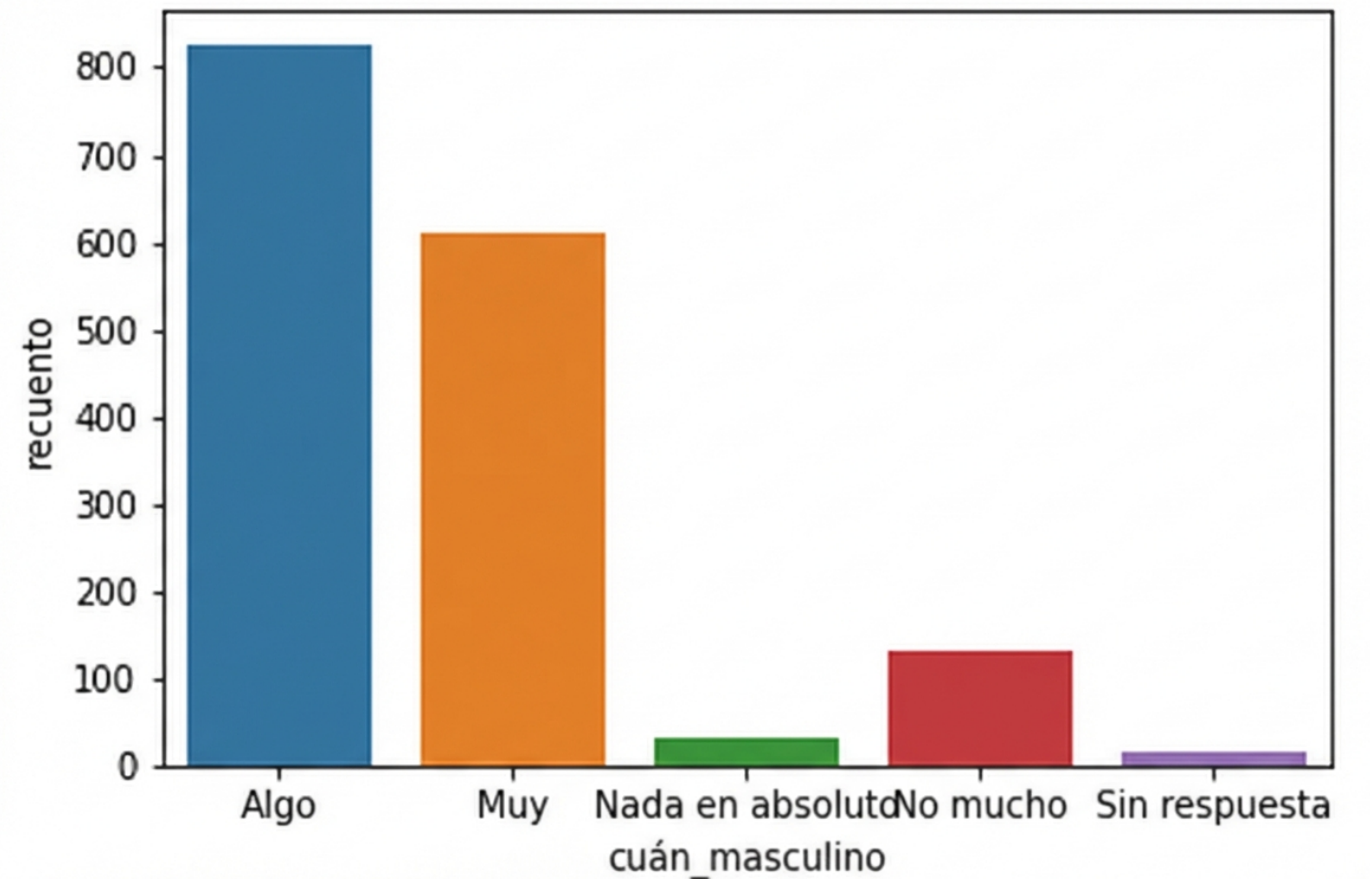
|   | participant_id | age     | how_masculine | how_important |
|---|----------------|---------|---------------|---------------|
| 0 | 1              | 18 - 34 | Somewhat      | Somewhat      |
| 1 | 2              | 18 - 34 | Somewhat      | Somewhat      |
| 2 | 3              | 18 - 34 | Very          | Not very      |
| 3 | 4              | 18 - 34 | Very          | Not very      |
| 4 | 5              | 18 - 34 | Very          | Very          |

# Usar DataFrames con countplot()

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("masculinity.csv")
sns.countplot(x="how_masculine",
              data=df)

plt.show()
```



|   | id_participante | edad    | cuán_masculino | cuán_importante |
|---|-----------------|---------|----------------|-----------------|
| 0 | 1               | 18 - 34 | Algo           | Algo            |
| 1 | 2               | 18 - 34 | Algo           | Algo            |
| 2 | 3               | 18 - 34 | Muy            | No mucho        |
| 3 | 4               | 18 - 34 | Muy            | No mucho        |
| 4 | 5               | 18 - 34 | Muy            | Muy             |
| 5 | 6               | 18 - 34 | Muy            | Algo            |
| 6 | 7               | 18 - 34 | Algo           | No mucho        |
| 7 | 8               | 18 - 34 | Algo           | Algo            |
| 8 | 9               | 18 - 34 | Muy            | Nada            |
| 9 | 10              | 18 - 34 | Algo           | Algo            |

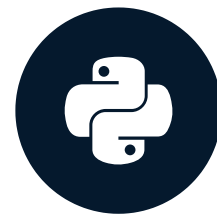
| ENTRE LOS HOMBRES ADULTOS |                                                                      |      |               |               |          |
|---------------------------|----------------------------------------------------------------------|------|---------------|---------------|----------|
| Sin nombre: 1             | Hombres adultos                                                      | Edad | Sin nombre: 4 | Sin nombre: 5 |          |
| 0                         |                                                                      |      | 18 - 34       | 35 - 64       | 65 y más |
| 1                         | En general, ¿cómo de masculino o 'varonil' te sientes?               |      |               |               |          |
|                           | Muy masculino                                                        | 37%  | 29%           | 42%           | 37%      |
|                           | Algo masculino                                                       | 46%  | 47%           | 46%           | 47%      |
|                           | Poco masculino                                                       | 11%  | 13%           | 9%            | 13%      |
|                           | Nada masculino                                                       | 5%   | 10%           | 2%            | 3%       |
|                           | Sin respuesta                                                        | 1%   | 0%            | 1%            | 1%       |
| 7                         | ¿Cómo de importante es para ti que los demás te vean como masculino? |      |               |               |          |
|                           | Muy importante                                                       | 16%  | 18%           | 17%           | 13%      |
|                           | Algo importante                                                      | 37%  | 38%           | 37%           | 32%      |
|                           | Poco importante                                                      | 28%  | 18%           | 31%           | 37%      |
|                           | Nada importante                                                      | 18%  | 26%           | 15%           | 18%      |
|                           | Sin respuesta                                                        | 0%   | 0%            | 1%            | 0%       |

# ¡Vamos a practicar!

INTRODUCCIÓN A LA VISUALIZACIÓN DE DATOS CON SEABORN

# Introducción a Seaborn

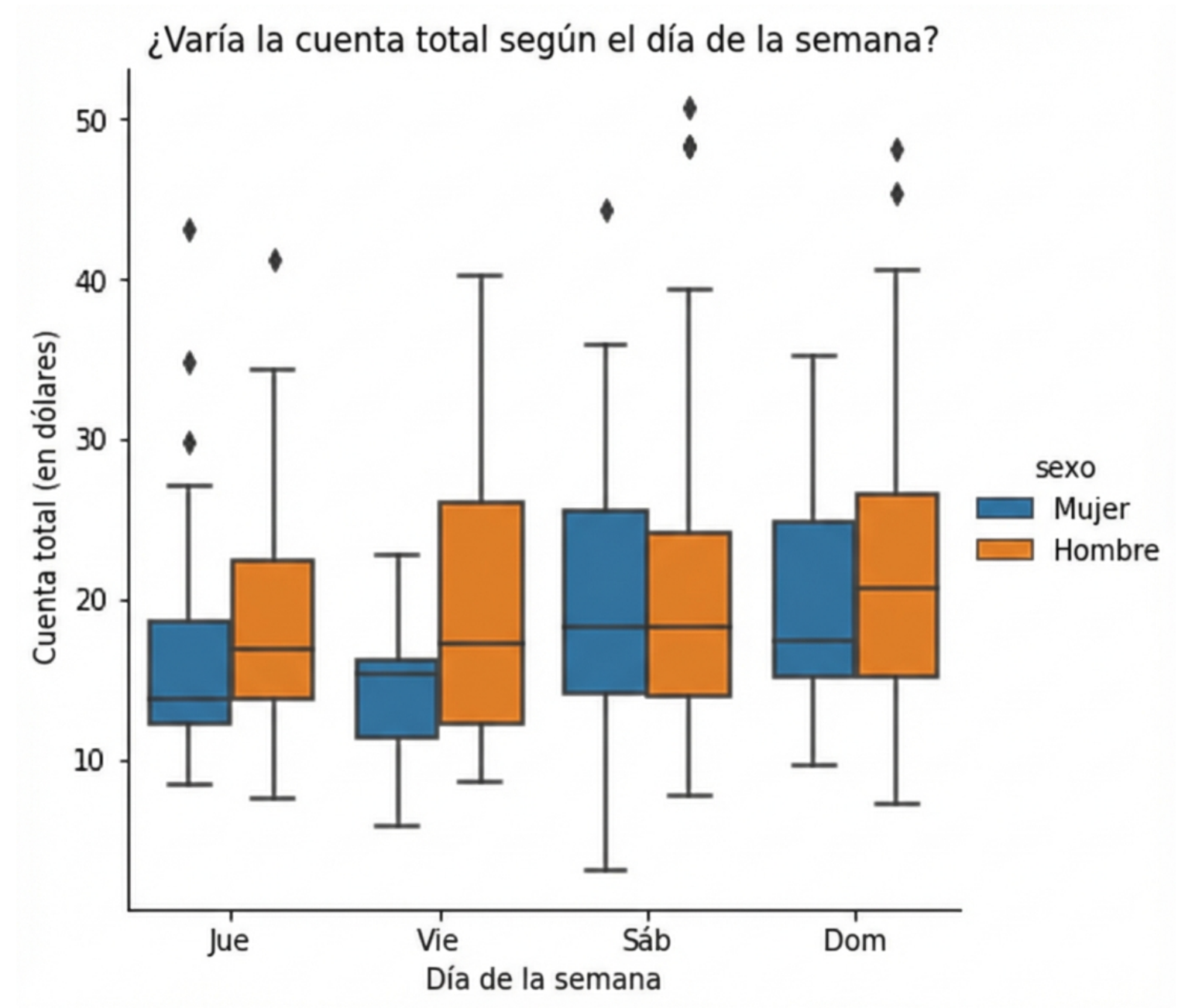
INTRODUCCIÓN A LA VISUALIZACIÓN DE DATOS CON SEABORN



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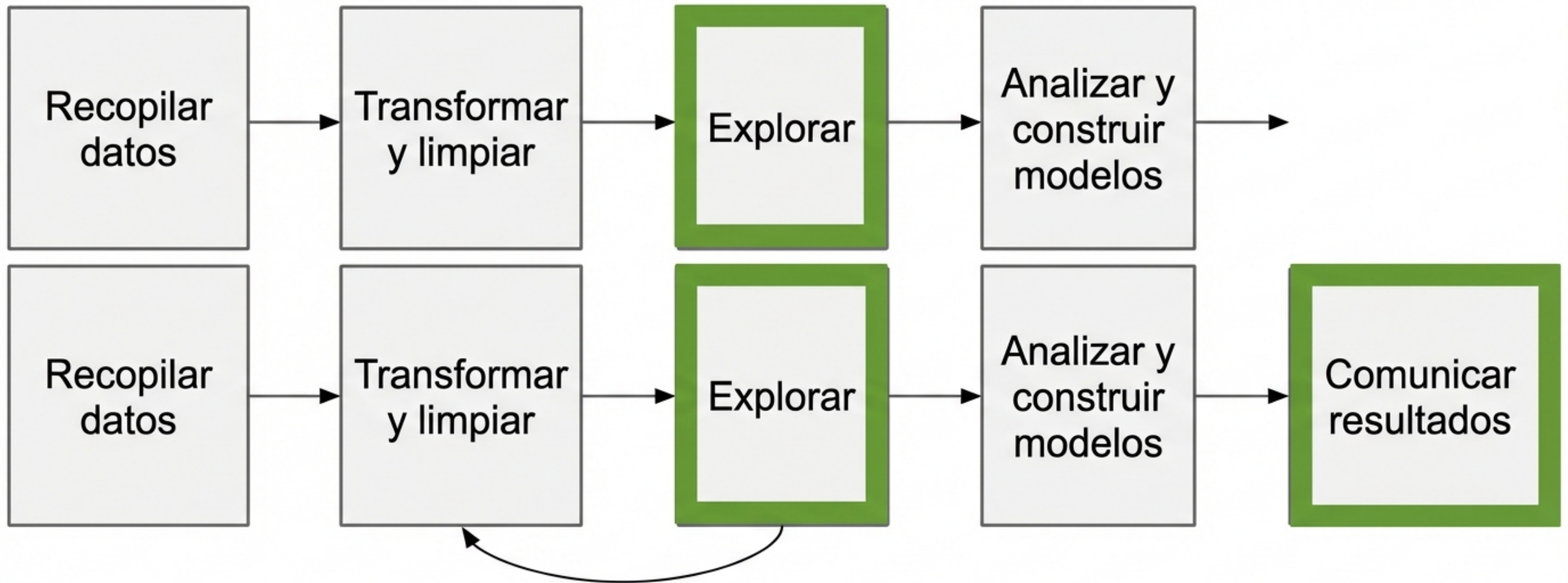
# ¿Qué es Seaborn?

- Biblioteca de visualización de datos de Python
- Crea fácilmente los tipos de gráficos más comunes



<sup>1</sup> Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

# ¿Por qué Seaborn es útil?



# Ventajas de Seaborn

- Fácil de usar
- Funciona bien con estructuras de datos de tipo `pandas`
- Se basa en `matplotlib`

# Empezamos

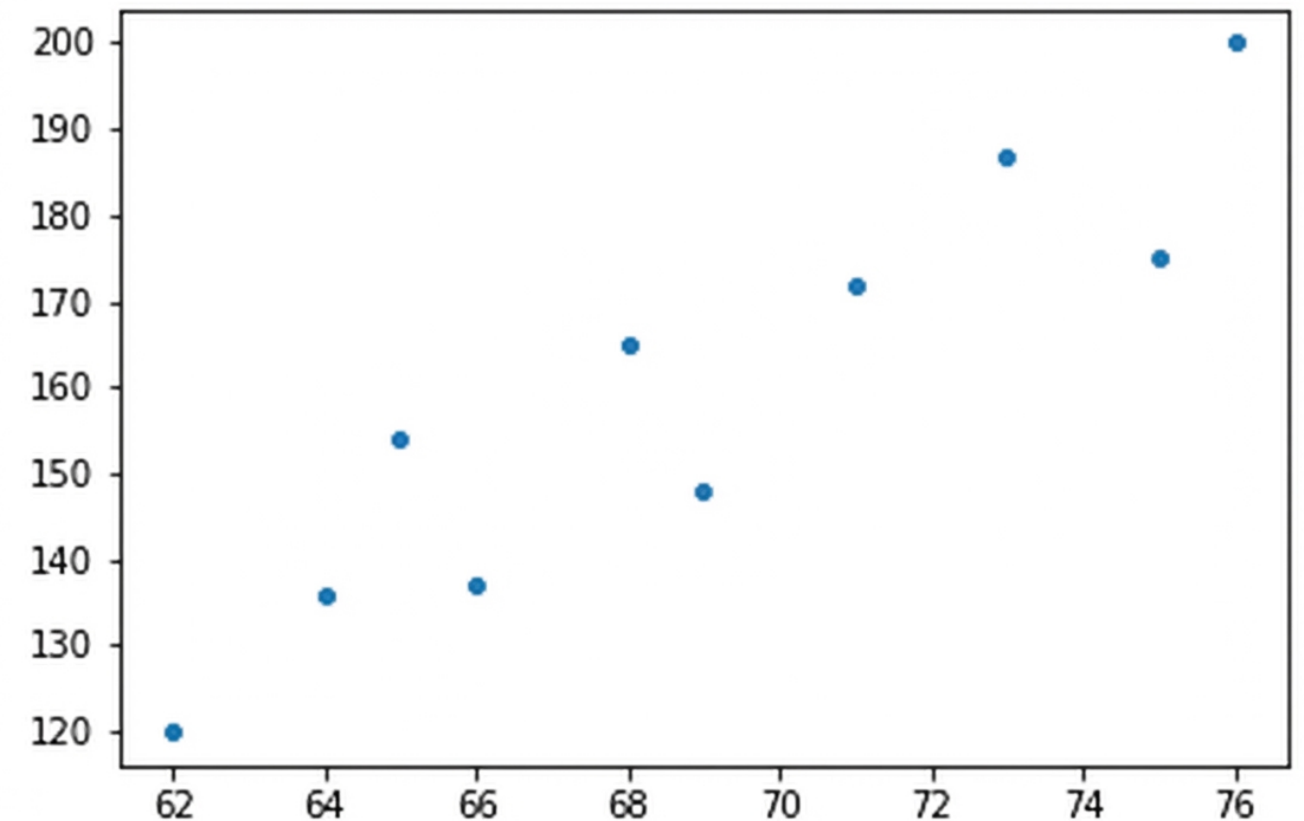
```
import seaborn as sns
import matplotlib.pyplot as plt
```

Samuel Norman Seaborn (sns)

- La serie de televisión «El ala oeste de la Casa Blanca»

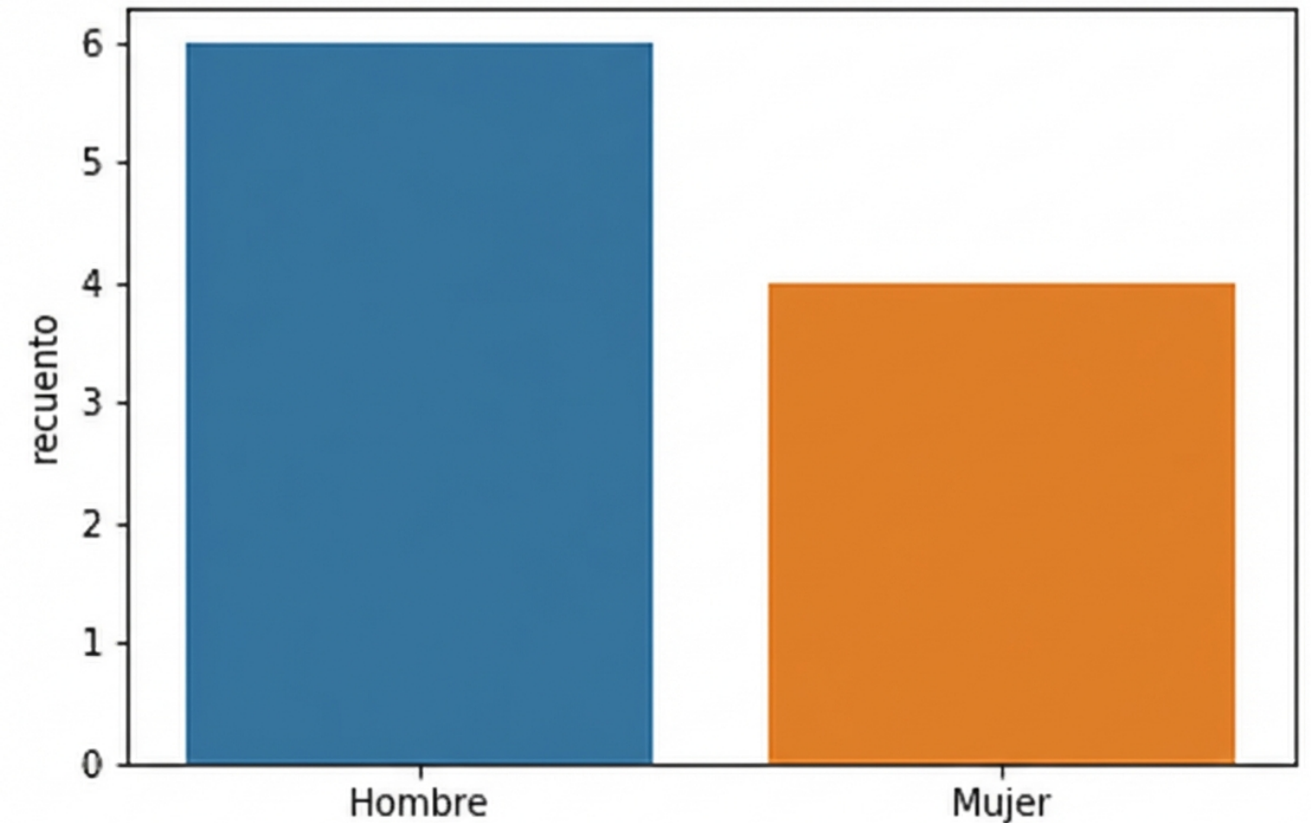
# Ejemplo 1: Diagrama de dispersión

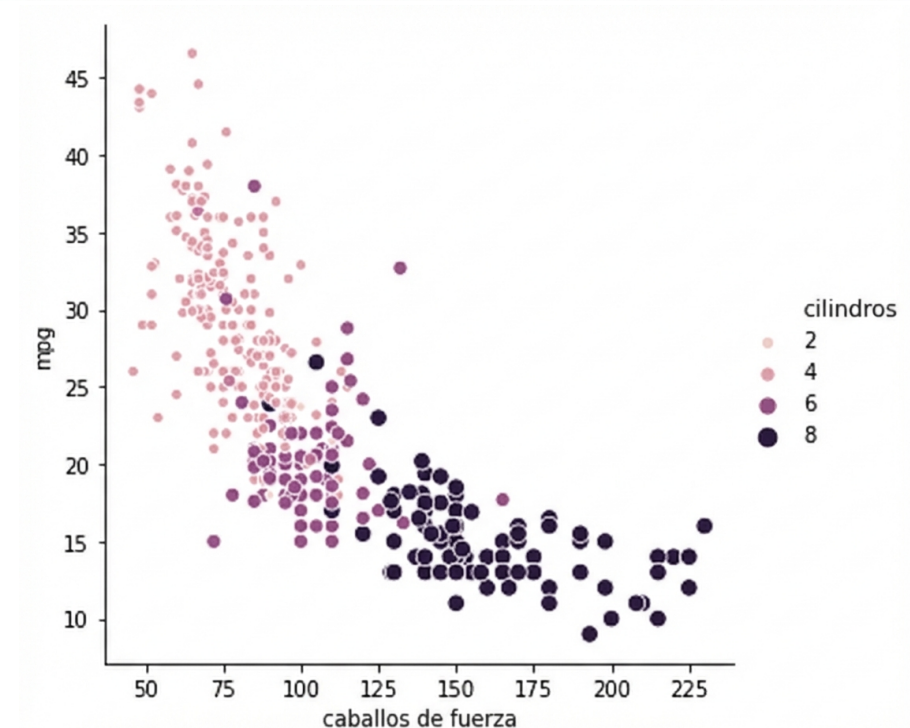
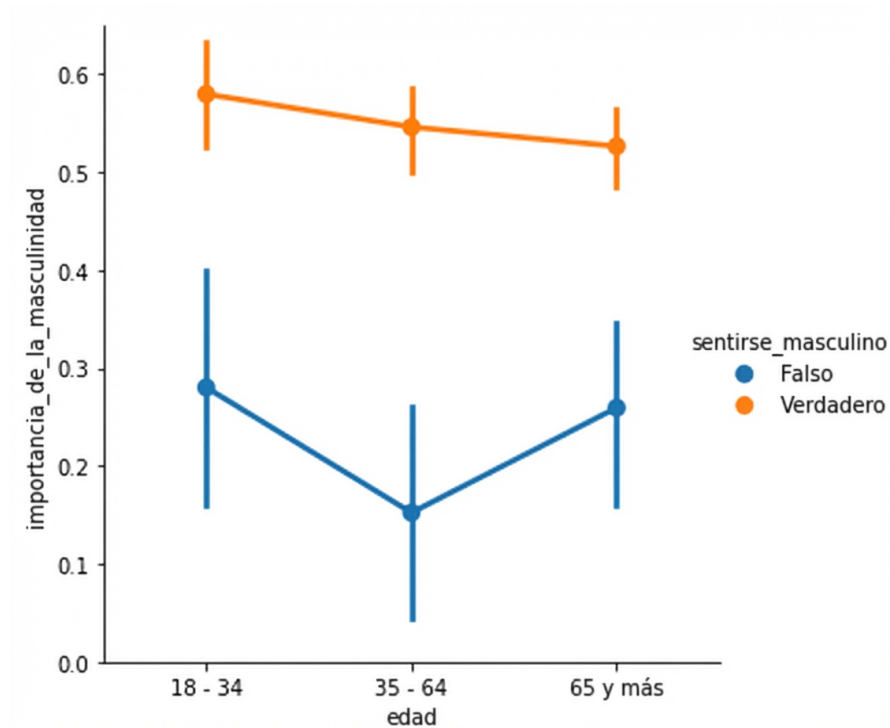
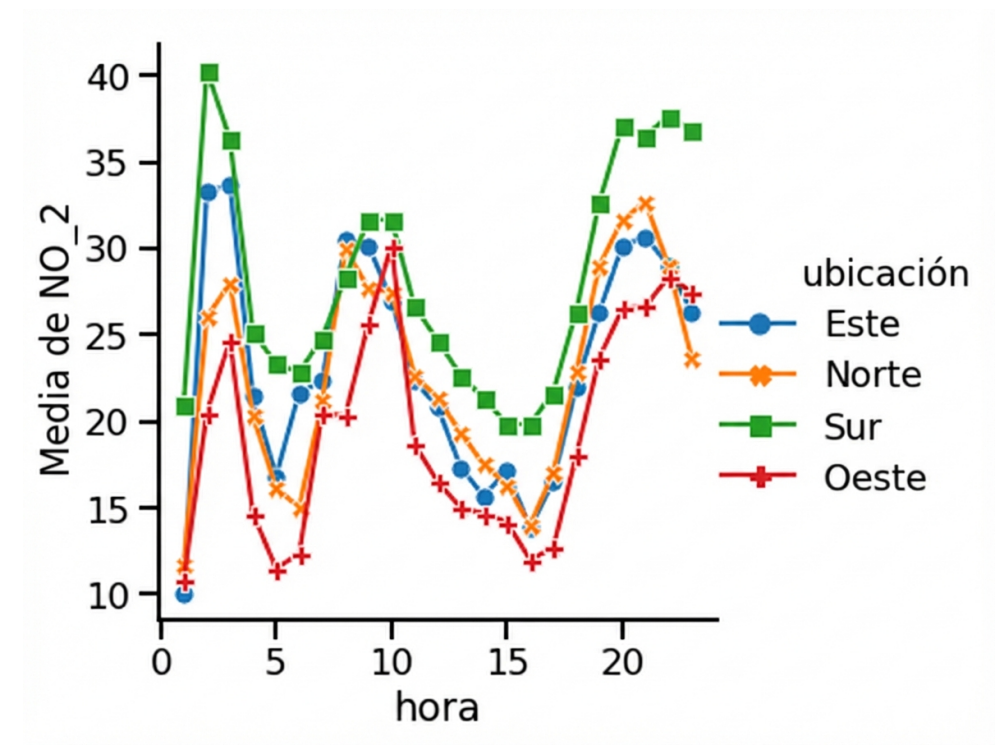
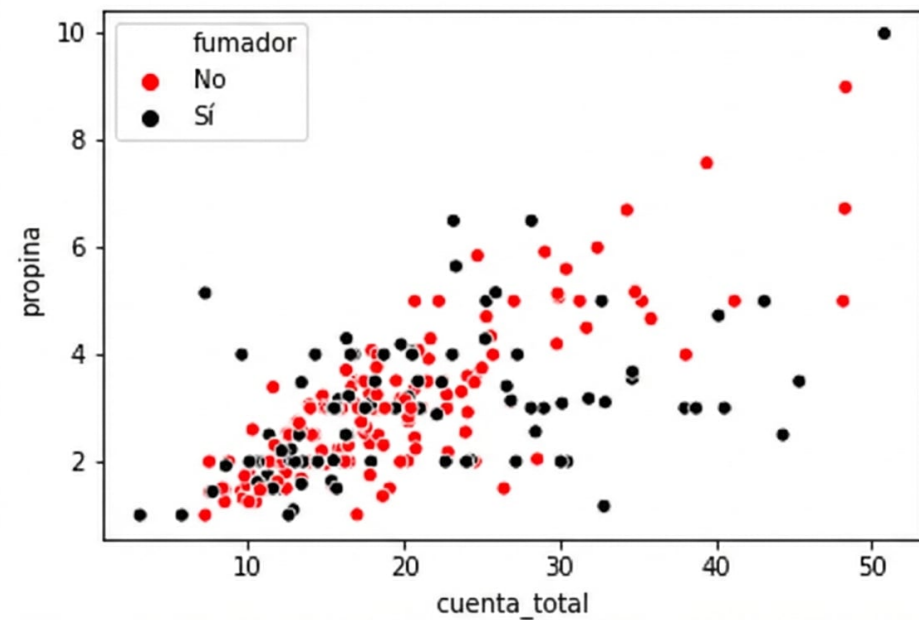
```
import seaborn as sns
import matplotlib.pyplot as plt
height = [62, 64, 69, 75, 66,
          68, 65, 71, 76, 73]
weight = [120, 136, 148, 175, 137,
          165, 154, 172, 200, 187]
sns.scatterplot(x=height, y=weight)
plt.show()
```



# Ejemplo 2: crear un gráfico de recuento

```
import seaborn as sns
import matplotlib.pyplot as plt
gender = ["Female", "Female",
          "Female", "Female",
          "Male", "Male", "Male",
          "Male", "Male", "Male"]
sns.countplot(x=gender)
plt.show()
```





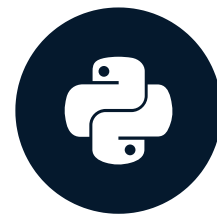
<sup>1</sup> Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

# ¡Vamos a practicar!

INTRODUCCIÓN A LA VISUALIZACIÓN DE DATOS CON SEABORN

# Añadir una tercera variable con el tono

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# Conjunto de datos de propinas

```
import pandas as pd
import seaborn as sns
tips = sns.load_dataset("tips")
tips.head()
```

|   | total_bill | tip  | sex    | smoker | day | time   | size |
|---|------------|------|--------|--------|-----|--------|------|
| 0 | 16.99      | 1.01 | Female | No     | Sun | Dinner | 2    |
| 1 | 10.34      | 1.66 | Male   | No     | Sun | Dinner | 3    |
| 2 | 21.01      | 3.50 | Male   | No     | Sun | Dinner | 3    |
| 3 | 23.68      | 3.31 | Male   | No     | Sun | Dinner | 2    |
| 4 | 24.59      | 3.61 | Female | No     | Sun | Dinner | 4    |

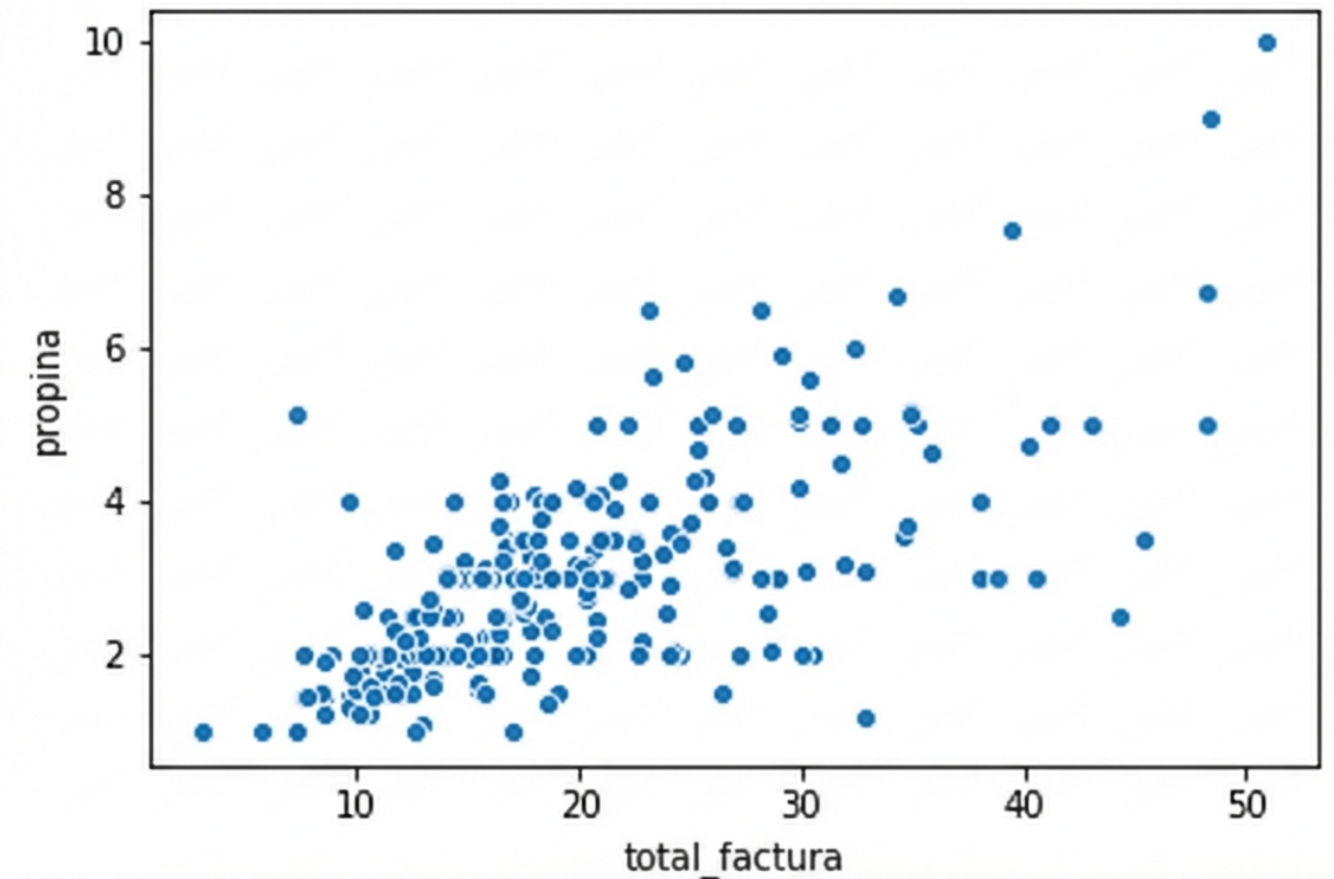
<sup>1</sup> Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

# Un gráfico de dispersión básico

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.scatterplot(x="total_bill",
                y="tip",
                data=tips)

plt.show()
```



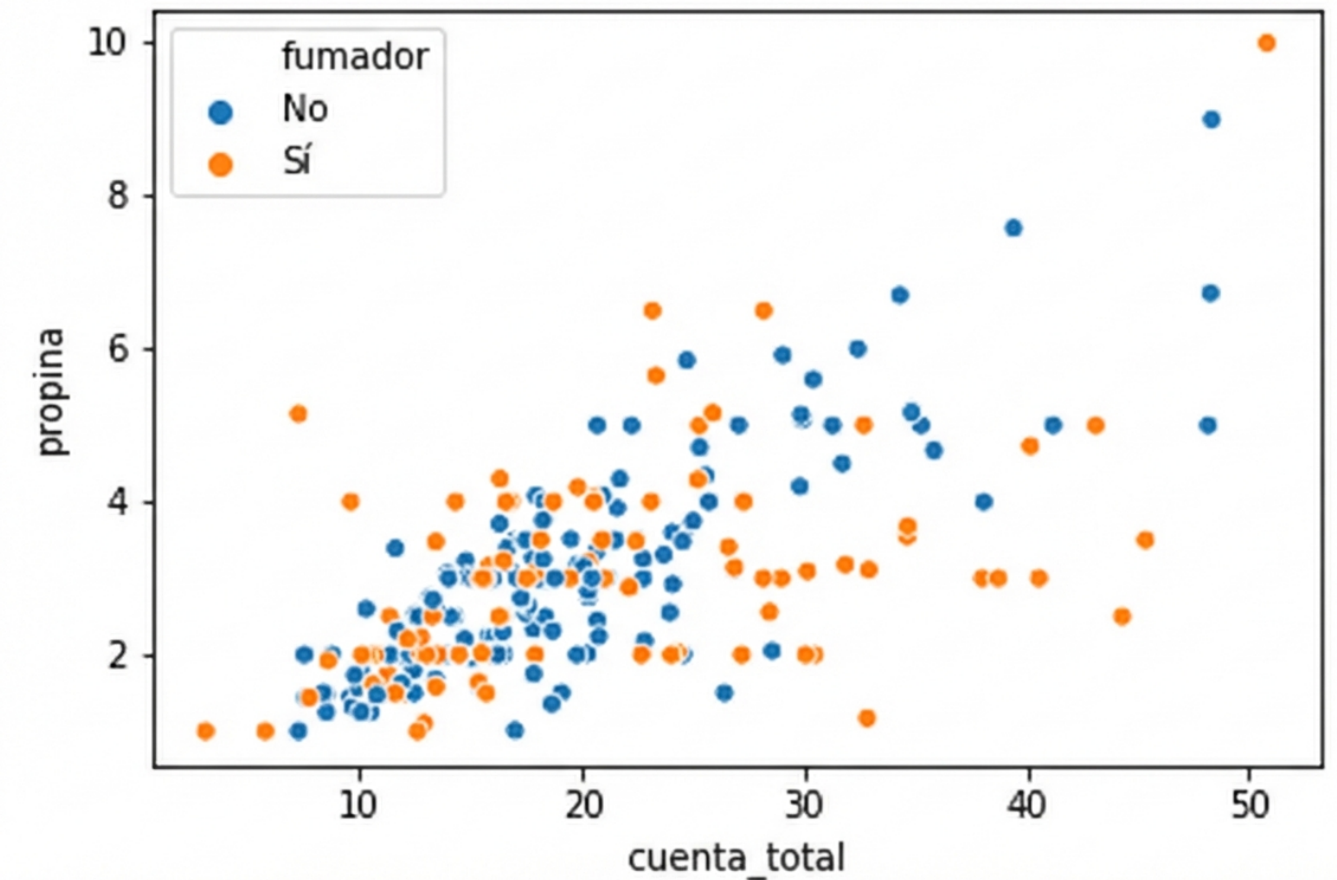
<sup>1</sup> Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

# Un gráfico de dispersión con tono

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.scatterplot(x="total_bill",
                y="tip",
                data=tips,
                hue="smoker")

plt.show()
```



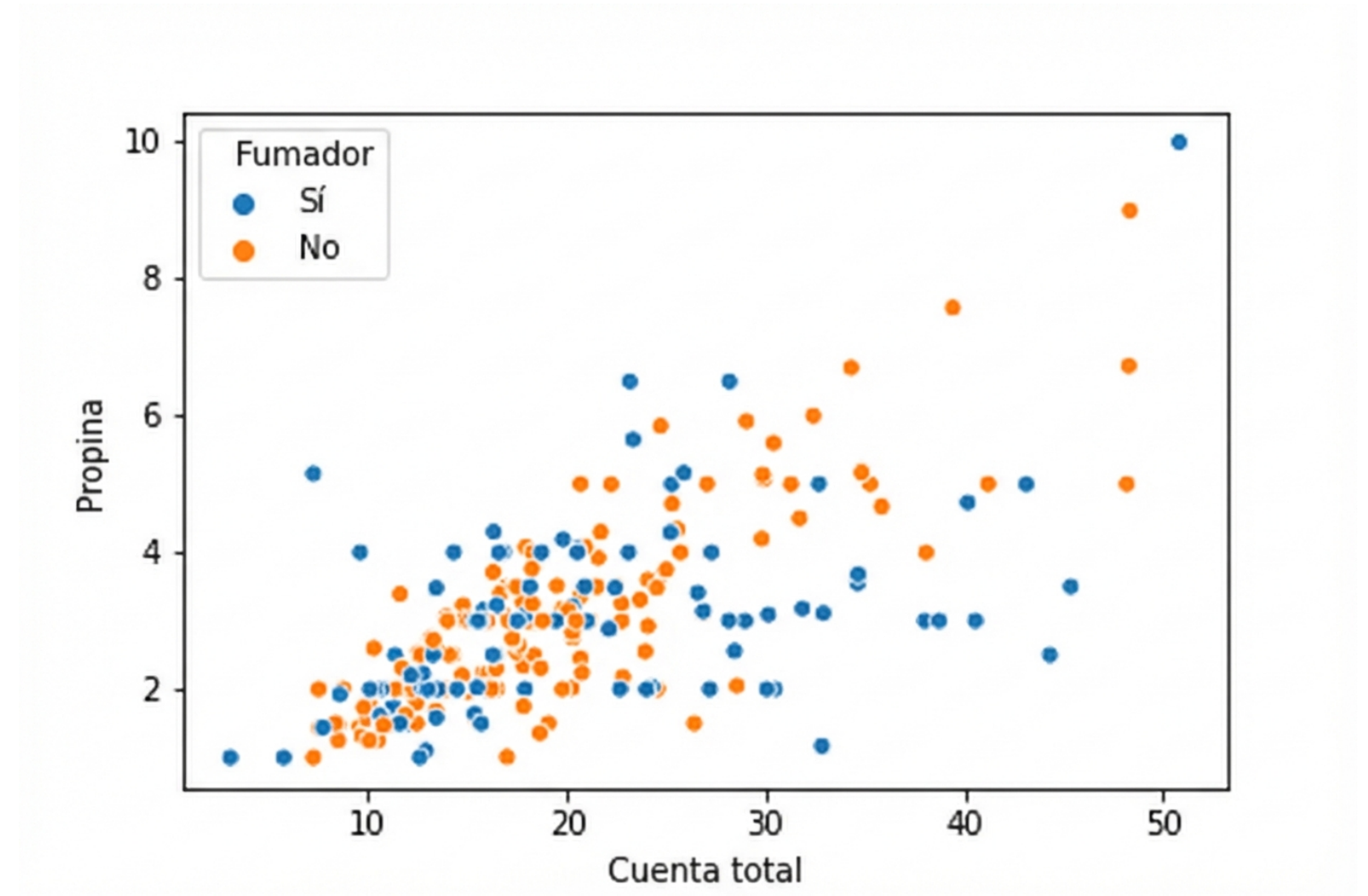
<sup>1</sup> Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

# Establecer el orden de los tonos

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.scatterplot(x="total_bill",
                y="tip",
                data=tips,
                hue="smoker",
                hue_order=["Yes",
                           "No"])

plt.show()
```



<sup>1</sup> Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

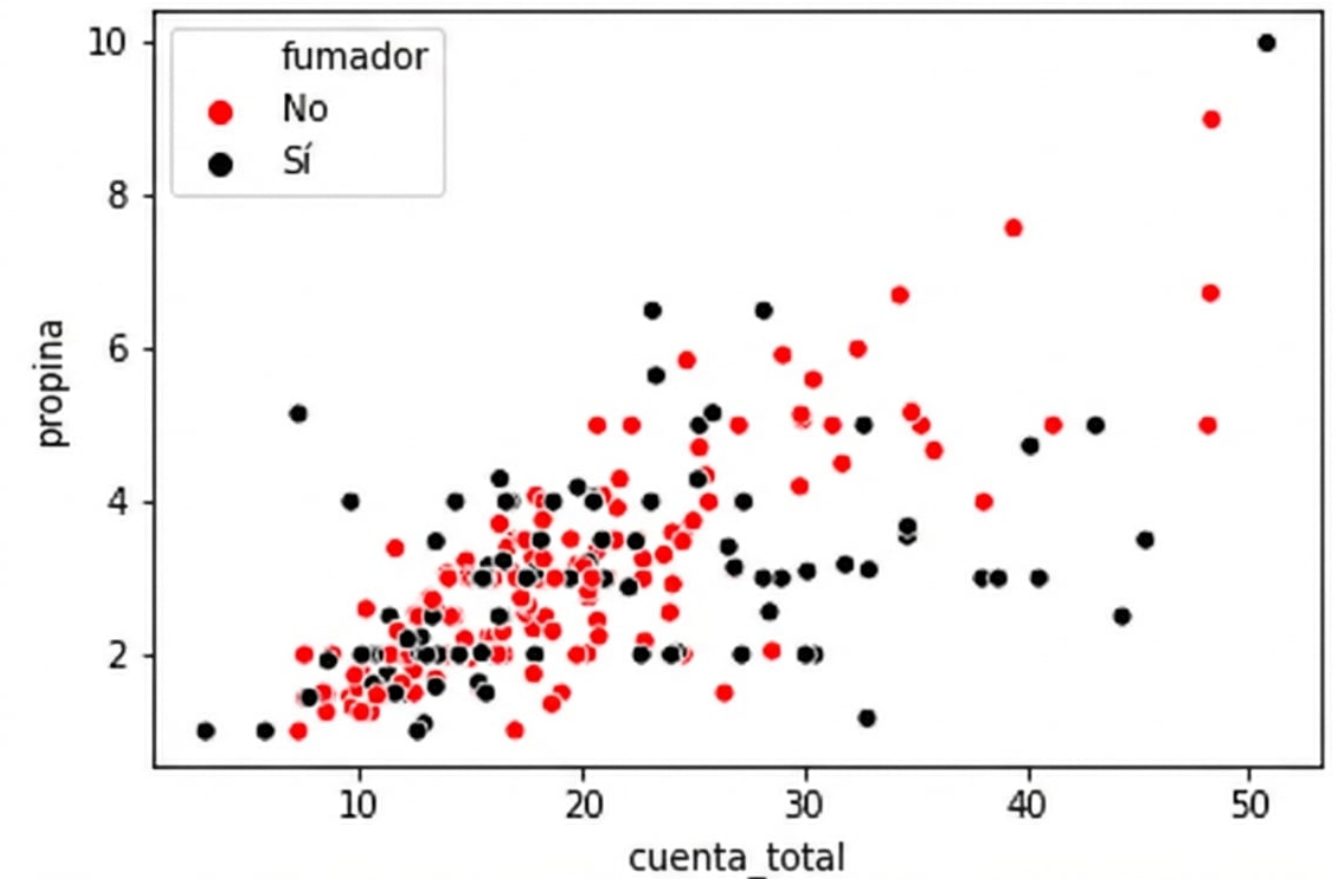
# Especificar colores de tono

```
import matplotlib.pyplot as plt
import seaborn as sns

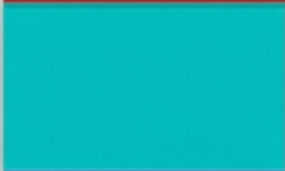
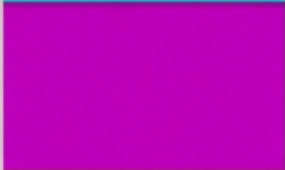
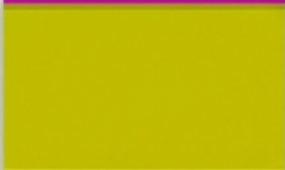
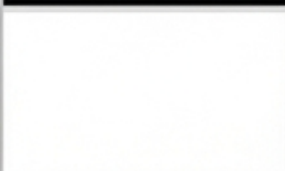
hue_colors = {"Yes": "black",
              "No": "red"}

sns.scatterplot(x="total_bill",
                y="tip",
                data=tips,
                hue="smoker",
                palette=hue_colors)

plt.show()
```



<sup>1</sup> Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

|                                                                                     | Color      | Nombre en Matplotlib | Abreviatura en Matplotlib | Código de color HTML (hex) |
|-------------------------------------------------------------------------------------|------------|----------------------|---------------------------|----------------------------|
|    | azul       | "blue"               | "b"                       | #0000ff                    |
|    | verde      | "green"              | "g"                       | #008000                    |
|    | rojo       | "red"                | "r"                       | #ff0000                    |
|    | verde/azul | "cyan"               | "c"                       | #00bfff                    |
|   | púrpura    | "magenta"            | "m"                       | #bf00bf                    |
|  | amarillo   | "yellow"             | "y"                       | #ffff00                    |
|  | negro      | "black"              | "k"                       | #000000                    |
|  | blanco     | "white"              | "w"                       | #ffffff                    |

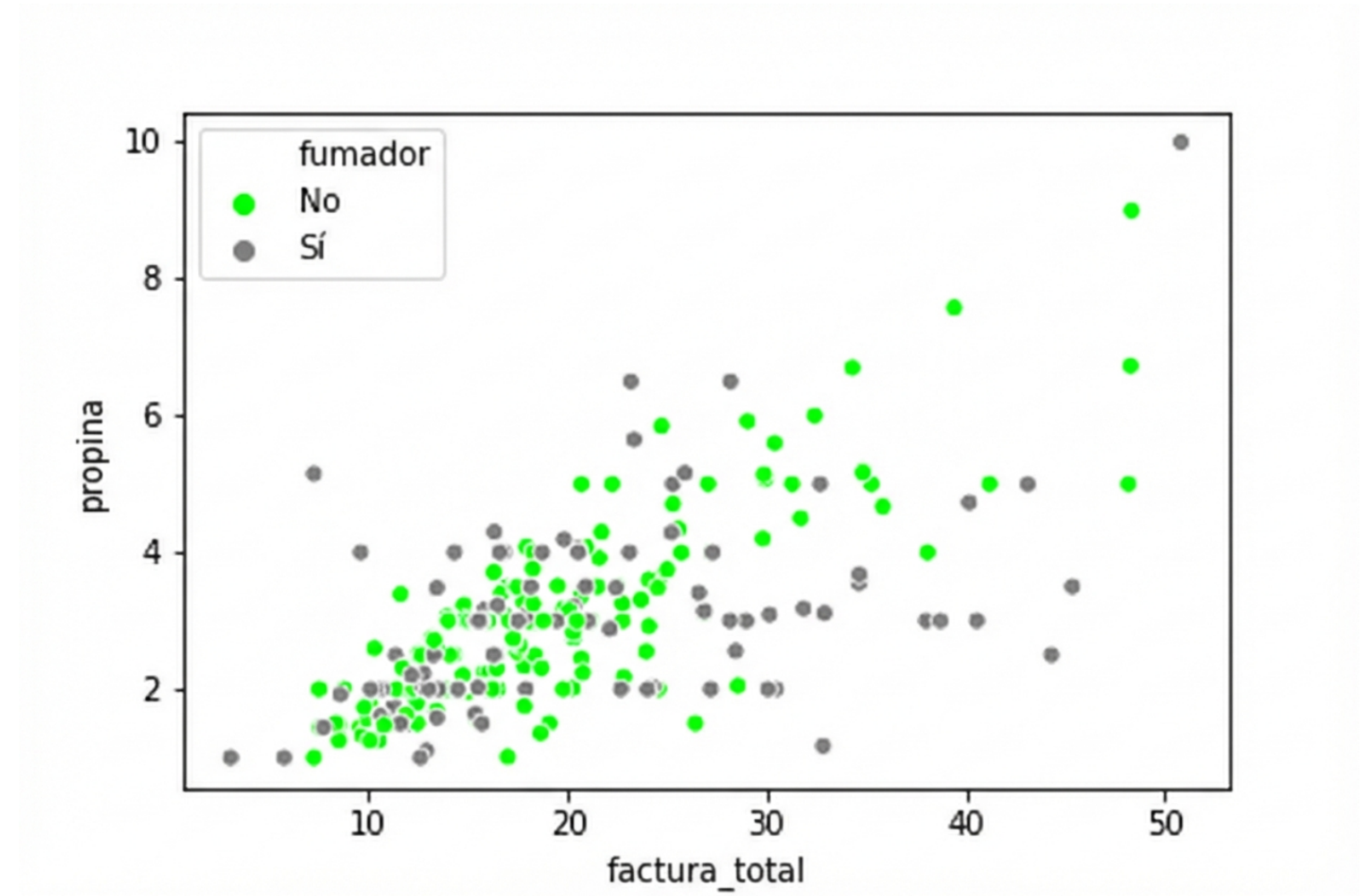
# Uso de códigos hexadecimales HTML con tono

```
import matplotlib.pyplot as plt
import seaborn as sns

hue_colors = {"Yes": "#808080",
              "No": "#00FF00"}

sns.scatterplot(x="total_bill",
                y="tip",
                data=tips,
                hue="smoker",
                palette=hue_colors)

plt.show()
```



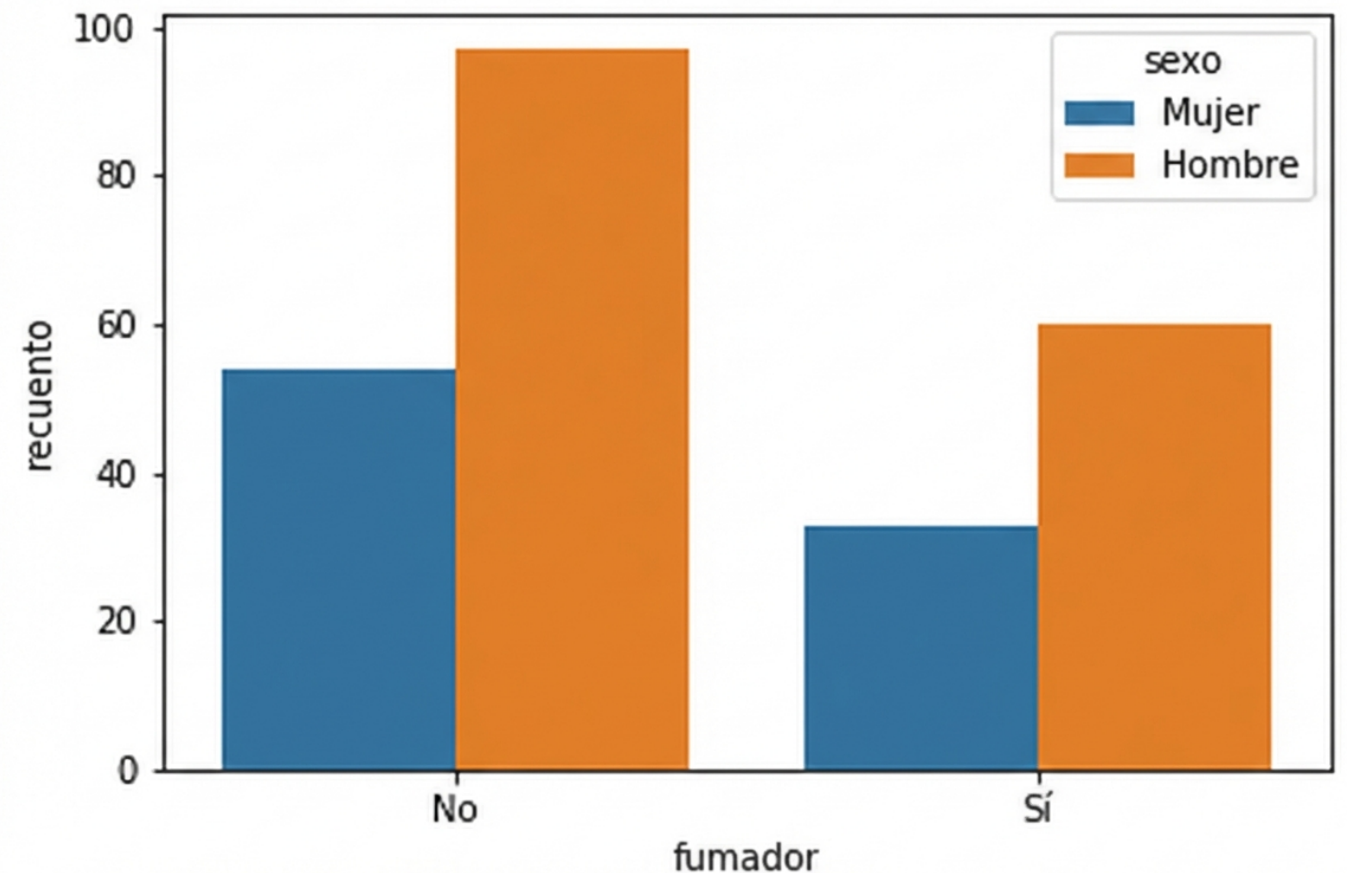
<sup>1</sup> Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

# Uso del tono con gráficos de recuento

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.countplot(x="smoker",
              data=tips,
              hue="sex")

plt.show()
```



<sup>1</sup> Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

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